EYE-CONIQ

AI-Powered Diabetic Retinopathy Screening Report

Report ID	EYC-20260829231630
Date & Time	August 29, 2026 • 11:16 PM
System	EYE-CONIQ v2.0 — Explainable AI DR Screening
Model	ResNet-18 (APTOS 2019 trained)

DIAGNOSIS

Moderate NPDR

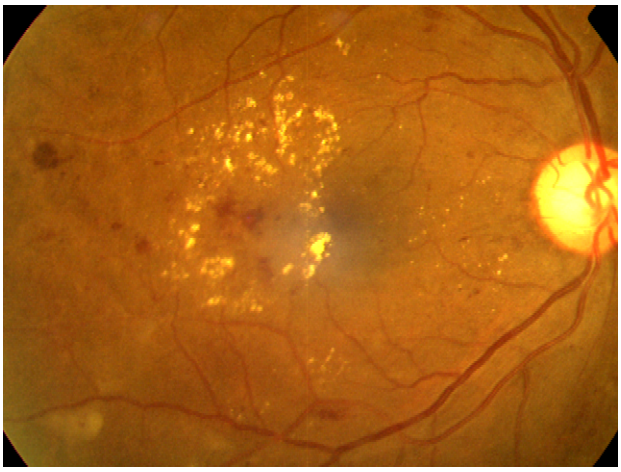
Level 2/4

Confidence: 55.4%

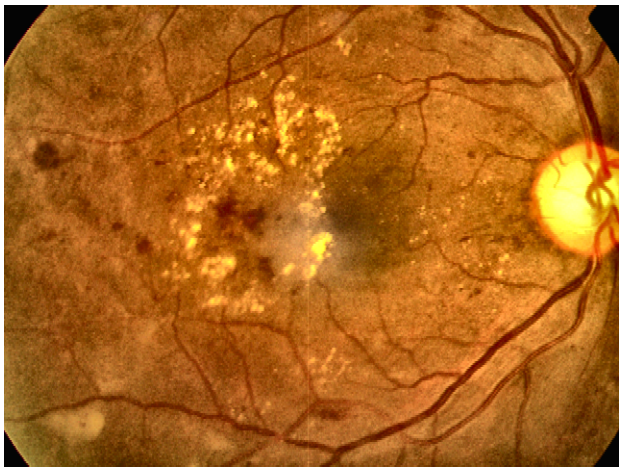
■ **Recommendation:** Refer to ophthalmologist within 3-6 months. Monitor for progression.

■ **REFERABLE DR DETECTED** (Probability: 65.5%)

RETINAL IMAGES

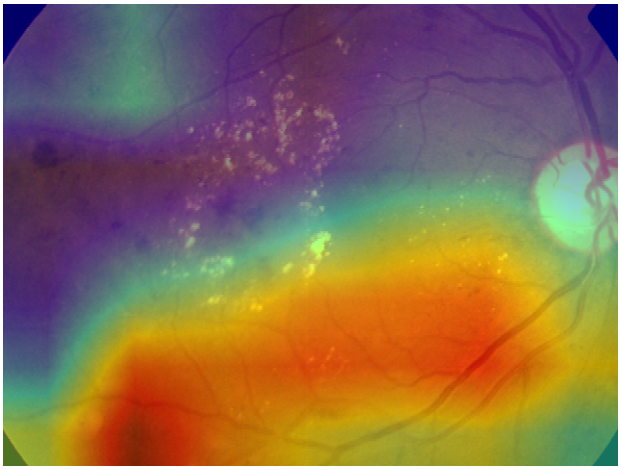


Original Image

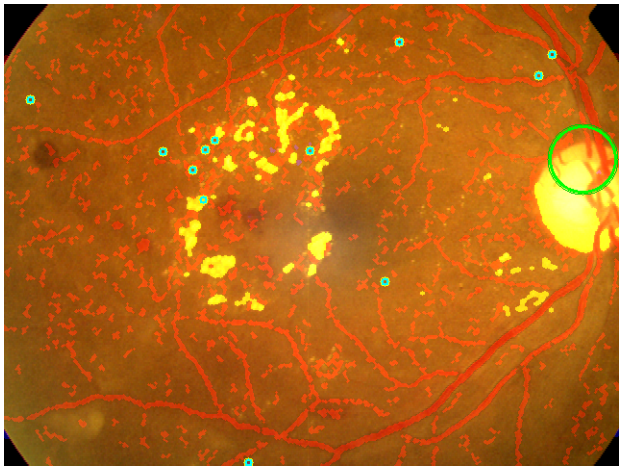


Enhanced Image

EXPLAINABILITY & STRUCTURE ANALYSIS



Grad-CAM Attention Map
Highlights regions influencing the AI decision



Retinal Structure Map
Red: Vessels | Yellow: Exudates | Blue: Hemorrhages

IMAGE QUALITY ASSESSMENT

Metric	Score	Status
Focus	68/100	■
Illumination	81/100	■
Field of View	72/100	■
Contrast	45/100	■■
Overall	68/100	Accept

LESION ANALYSIS

Structure	Count / Value
Microaneurysms	12
Exudate Regions	47
Hemorrhage Regions	8
Vessel Density	16.6%
Optic Disc Detected	Yes

SPATIAL ATTENTION ANALYSIS

Most affected quadrant: Inferior Nasal
Activation coverage: 51.0% of retina
Peak activation: 1.00
Model uncertainty: 61.7%

Quadrant	Activation Level
Superior Temporal	■■■■■■■■■■■■■■■■■■■■■■ 0.151
Superior Nasal	■■■■■■■■■■■■■■■■■■■■■■ 0.156
Inferior Temporal	■■■■■■■■■■■■■■■■■■■■■■ 0.469
Inferior Nasal	■■■■■■■■■■■■■■■■■■■■■■ 0.622

CLASS PROBABILITY DISTRIBUTION

DR Level	Probability
No DR	■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ 0.0%
Mild NPDR	■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ 34.4%
Moderate NPDR	■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ 55.4%
Severe NPDR	■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ 6.7%
Proliferative DR	■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ 3.4%

DISCLAIMER: This report is generated by an AI-assisted screening system and is intended to support — not replace — clinical judgment. All findings must be reviewed and confirmed by a qualified ophthalmologist before any treatment decisions. This system has been designed for primary screening in resource-limited settings and should be used as part of a human-in-the-loop diagnostic workflow.